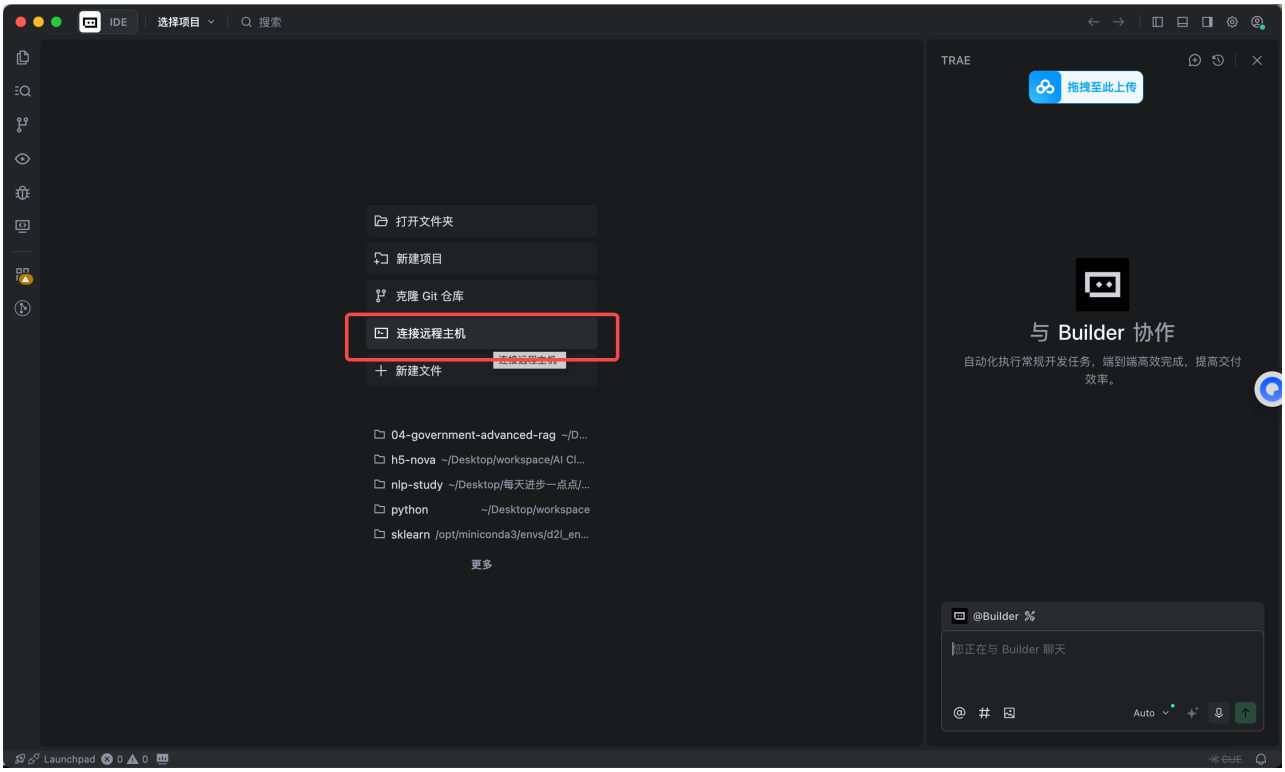
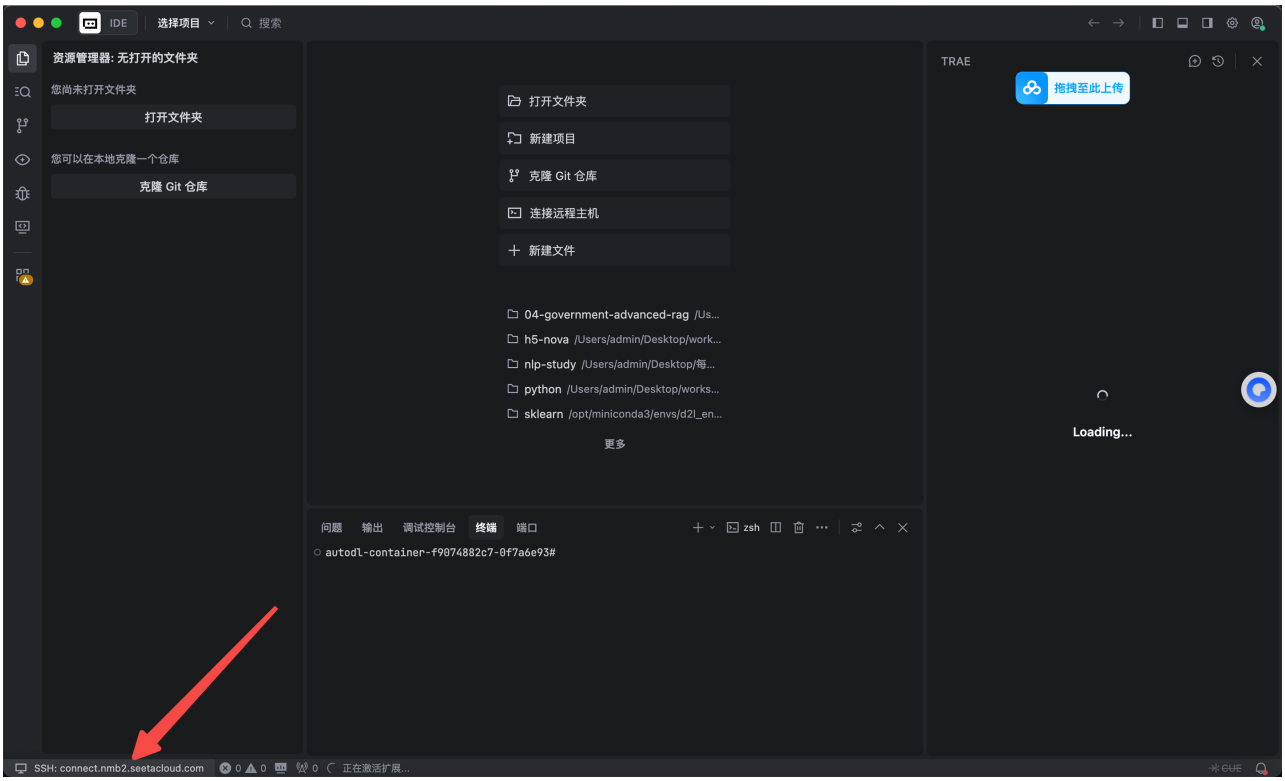
- 第一步：租gpu服务器(RTX 3090级别，24GB显存)https://autodl.com/console/instance/list，镜像选择ptorch
- 第二步：将本地代码上传到服务器

打开编辑器，选择连接远程主机

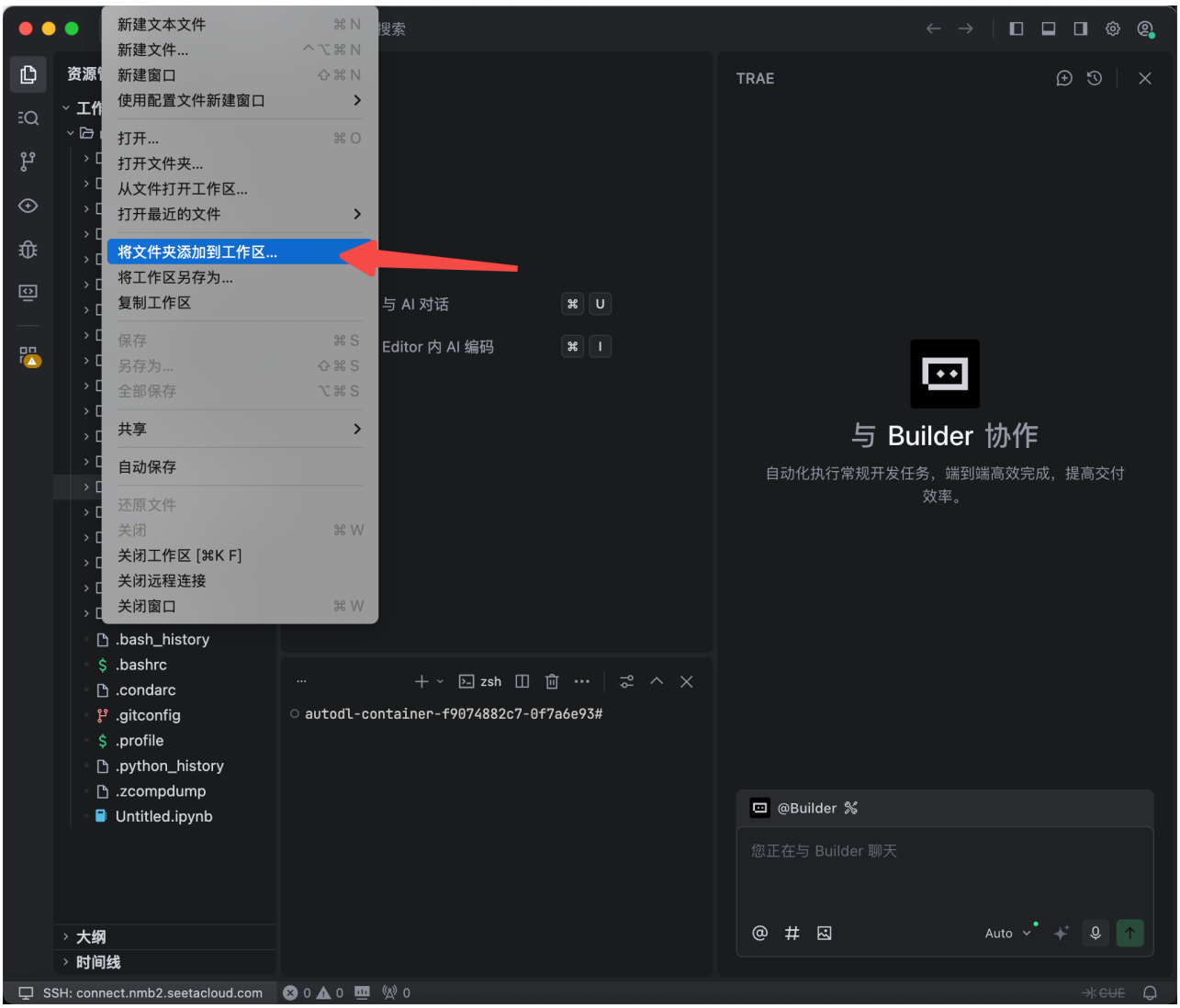


输入登录指令和密码，即可连接服务器

表示连接成功



将文件添加到工作区，添加完之后，把本地代码直接拖到对应目录下即可。



- 第三步，安装vllm pip install vllm

```
2.清理系统盘请参考：https://www.autodl.com/docs/qa1/
3.终端中长期执行命令请使用screen等工具开后台运行，确保程序不受SSH连接中断影响：https://www.autodl.com/docs/da
root@autodl-container-f9074882c7-0f7a6e93:~# vllm
usage: vllm [-h] [-v] {chat,complete,serve,bench,collect-env,run-batch} ...

vLLM CLI

positional arguments:
  {chat,complete,serve,bench,collect-env,run-batch}
    chat                Generate chat completions via the running API server.
    complete            Generate text completions based on the given prompt via the running API server.
    serve               Launch a local OpenAI-compatible API server to serve LLM completions via HTTP.
    bench               vLLM bench subcommand.
    collect-env         Start collecting environment information.
    run-batch           Run batch prompts and write results to file.

options:
  -h, --help            show this help message and exit
  -v, --version         show program's version number and exit

For full list:            vllm [subcommand] --help=all
For a section:            vllm [subcommand] --help=ModelConfig    (case-insensitive)
For a flag:               vllm [subcommand] --help=max-model-len  (_ or - accepted)
Documentation:            https://docs.vllm.ai
root@autodl-container-f9074882c7-0f7a6e93:~#
```

- 第四步，使用vllm部署本地大模型

```
pip install modelscope
# 创建模型目录
mkdir -p /root/autodl-tmp/models/Qwen2.5-7B-Instruct
# 使用 modelscope-cli 下载（或者手动下载后放入该目录）
# 这里演示使用 Python 脚本快速下载
python -c "from modelscope import snapshot_download; snapshot_download('Qwen/Qwen2.5-7B-Instruct', cache_dir='/root/autodl-tmp/models')"
```

下载完成之后启动vLLM服务

```
python -m vllm.entrypoints.openai.api_server \
    --model /root/models/Qwen/Qwen2.5-7B-Instruct \
    --host 0.0.0.0 \
    --port 8000 \
    --dtype auto \
    --max-model-len 8192
```

- 第五步，安装mysql,数据持久化，允许远程连接（未完成）

```
apt-get update
apt-get install -y mysql-server
# 在持久化数据盘创建目录
mkdir -p /root/autodl-tmp/mysql_data

# 赋予正确的权限（mysql 用户需要能读写）
chown -R mysql:mysql /root/autodl-tmp/mysql_data
chmod -R 750 /root/autodl-tmp/mysql_data

```

启动成功：

```
root@autodl-container-f9074882c7-0f7a6e93:~# service mysql start
 * Starting MySQL database server mysqld
root@autodl-container-f9074882c7-0f7a6e93:~# mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.45-0ubuntu0.22.04.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

- 第六步安装es等，进到项目里面，安装依赖

pip install -r requirements.txt

- 启动项目

```
FileNotFoundError: Path /root/models/BAAI/bge-small-zh-v1.5/ not found
root@autodl-container-f9074882c7-0f7a6e93:~/04-government-advanced-rag# python main.py
Could not connect to Elasticsearch.
Loading embedding model bge-small-zh-v1.5 from model_path...
2026-03-20 00:24:02,062 INFO sqlalchemy.engine.Engine BEGIN (implicit)
2026-03-20 00:24:02,062 INFO sqlalchemy.engine.Engine PRAGMA main.table_info("knowledge_database")
2026-03-20 00:24:02,062 INFO sqlalchemy.engine.Engine [raw sql] ()
2026-03-20 00:24:02,062 INFO sqlalchemy.engine.Engine PRAGMA main.table_info("knowledge_document")
2026-03-20 00:24:02,062 INFO sqlalchemy.engine.Engine [raw sql] ()
2026-03-20 00:24:02,063 INFO sqlalchemy.engine.Engine COMMIT
INFO:     Started server process [71617]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     Uvicorn running on http://0.0.0.0:6010 (Press CTRL+C to quit)
```

0.0.0.0:6010/docs#/default/get_knowledge_base_v1_knowledge_base_get

default

GET /v1/knowledge_base Get Knowledge Base

Parameters

Name	Description
knowledge_id * required integer (query)	knowledge_id
token * required string (query)	token

Responses

Code	Description
200	Successful Response

Media type

application/json

Controls Accept header.

Example Value | Schema


```
{
  "request_id": "string",
  "knowledge_id": 0,
  "category": "string",
  "title": "string",
  "response_code": 0,
  "response_msg": "string",
```